FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE II: FREE ELECTRON THEORY AND BAND THEORY

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

Bloch functions

1 What is a Bloch function? Write its form in terms of a periodic function $u_k(x)$. **(1 mark)**

Fermi function

2 Write the Fermi–Dirac occupation $f(E)$ at 0 K for EE_{F0}. **(1 mark)**

Velocity in a band

3 Write the group velocity of an electron in a band in terms of $E(k)$. **(1 mark)**

Number of states in a band

4 How many electrons can a single band of a crystal of N primitive cells hold? **(1 mark)**

Section B: 2 marks each, answer all

Free electron gas, 3D

5 Write the number of electrons N of a three-dimensional free electron gas in terms of the volume V and the Fermi energy E_F at 0 K. **(2 marks)**

Fermi temperature

6 Define the Fermi temperature T_F and give its relation with the Fermi energy. **(2 marks)**

Average kinetic energy

7 State the relation between the average kinetic energy per electron and the Fermi energy E_{F0} for a three-dimensional free electron gas at 0 K. **(2 marks)**

Free electron gas, 1D box

8 For a free electron gas in a one-dimensional box of length L with N electrons, show that the top filled level gives $E_F = \dfrac{\hbar^2}{2m}\left(\dfrac{N\pi}{2L}\right)^2$. **(2 marks)**

Section C: 4 marks each, answer all

Density of states

9 Show that the density of states of a three-dimensional free electron gas is $D(E) = \dfrac{V}{2\pi^2}\left(\dfrac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$. Count the states inside a sphere of radius k in k-space. **(4 marks)**

Fermi function

10 Show that the Fermi function equals $\tfrac{1}{2}$ at $E = E_F$ for any temperature, and explain why only electrons within about $k_B T$ of E_F are excited. **(4 marks)**

Effective mass

11 Show that the effective mass of an electron in a band is $m^* = \hbar^2/(d^2E/dk^2)$, starting from the acceleration produced by a field. **(4 marks)**

12 For $E = -E_0 \cos ka$, show that the effective mass at $k = 0$ is $m^* = \hbar^2/(E_0 a^2)$. **(4 marks)**

Filled band and conduction

13 Explain why a completely filled band has zero effective number of free electrons, using $N_{\text{eff}} = \dfrac{2Lm}{\pi\hbar^2}\left(\dfrac{dE}{dk}\right)_{k_1}$. **(4 marks)**

Kronig–Penney model

14 State the Kronig–Penney condition $P\dfrac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$. Give the limits $P \to 0$ and $P \to \infty$ and what each gives. **(4 marks)**

Section D: 10 marks each, answer any ONE

Free electron gas, 3D

15 Derive the Fermi energy E_{F0}, the average kinetic energy $E_0 = \tfrac{3}{5}E_{F0}$, and the density of states $D(E)$ of a three-dimensional free electron gas at 0 K. **(10 marks)**

Kronig–Penney model

16 Derive the Kronig–Penney dispersion relation $P\frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$ in the delta-function limit, state how allowed and forbidden bands follow from it, and discuss the limits of large and small P. **(10 marks)**

Metals, insulators and semiconductors

17 Using the band picture, distinguish between a metal, an intrinsic semiconductor and an insulator. Show that the effective number of free electrons in a completely filled band is zero and that it is largest when the band is filled up to the point of inflexion. **(10 marks)**

MODULE II: MODEL ANSWERS

For checking your work. Each answer shows the key step and the result.

Section A. 1. $u_k(x)$ is periodic with the lattice period, and $\psi(x) = e^{ikx}u_k(x)$ (book Eq. 6.4). **2.** $f(E) = 1$ for EE_{F0} (book Eq. 5.29). **3.** $v_g = \frac{1}{\hbar}\frac{dE}{dk}$ (book Eq. 6.49). **4.** $2N$ electrons (two spins in each of the N k-states of the band; book §6.3.2).

Section B. 5. $N = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2}$ (book Eq. 5.37). **6.** $T_F = E_F/k_B$ (book §5.3.1). **7.** E_{F0} and E_0 are related by $E_0 = \frac{3}{5}E_{F0}$ (book §5.2.2). **8.** $E_n = n^2h^2/8mL^2$; the top level $n_F = N/2$ gives $E_F = \frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2$ (book Eqs. 5.10, 5.12, 5.13).

Section C. 9. Count the k-states in a sphere of radius k: $2 \cdot \frac{4}{3}\pi k^3/(2\pi/L)^3 = \frac{L^3}{3\pi^2}k^3 = N$. With $k = \sqrt{2mE}/\hbar$ and $N(E) = \frac{V}{3\pi^2}\left(\frac{2mE}{\hbar^2}\right)^{3/2}$, differentiate: $D(E) = dN/dE$ gives Eq. 5.40. **10.** At $E = E_F$, $f = 1/[e^0 + 1] = \frac{1}{2}$ for any T. At $T > 0$ electrons within about k_BT of E_F can jump to empty states; those deeper cannot (book §5.2.2). **11.** $a = \frac{e\mathcal{E}}{\hbar^2}\frac{d^2E}{dk^2}$ (book Eq. 6.53); comparing with $a = e\mathcal{E}/m$ gives $m^* = \hbar^2/(d^2E/dk^2)$ (Eq. 6.55). **12.** $d^2E/dk^2 = E_0a^2\cos ka$, which is E_0a^2 at $k = 0$. So $m^* = \hbar^2/(E_0a^2)$. **13.** At a completely filled band $k_1 = \pi/a$ and $dE/dk = 0$ there, so $N_{\text{eff}} = 0$ (book Eq. 6.59). N_{eff} is largest when dE/dk is largest, at the inflexion point. **14.** Condition (6.39). $P \to 0$: $\cos\alpha a = \cos ka$, so $\alpha = k$, the free electron case with all energies allowed. $P \to \infty$: $\sin\alpha a = 0$, so $\alpha a = \pm n\pi$, which gives levels (book §6.3).

Section D. 15. $E_{F0} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$ (5.47) from $N = \int_0^{E_{F0}} D\,dE$; $E_0 = \frac{1}{N}\int_0^{E_{F0}} E\,D\,dE = \frac{3}{5}E_{F0}$; $D(E)$ as Eq. (5.40). **16.** Bloch form and the Wronskian give $\lambda^2 - (\alpha_1 + \beta_2)\lambda + 1 = 0$; in the delta limit (6.37) with P (6.38), giving Eq. (6.39). Allowed bands where the left side lies in $[-1, 1]$; forbidden otherwise (Fig. 6.3). Limits as C14. **17.** Metal: partly filled band, $N_{\text{eff}} \neq 0$. Insulator: full bands with a wide gap (diamond about 6 eV). Intrinsic semiconductor: full band with a narrow gap (about 1 eV), conduction by thermal excitation. Full band: $N_{\text{eff}} = 0$ (6.59); maximum at the inflexion point.

Mapping to the syllabus and textbook

Syllabus item	Topic	Puri and Babbar section	Model questions
6	Drude–Lorentz classical theory	5.1	none (covered by the past papers)
7	Sommerfeld theory, one-dimensional box	5.2.1 (Eqs. 5.10–5.13)	8
8	Fermi energy, total energy, density of states	5.2.2, 5.3	2, 5, 9, 15
9	Application of the free electron gas model	5.3, 5.3.1	6, 7
10	Bloch theorem, Kronig–Penney, velocity and effective mass	6.1–6.4	1, 3, 11, 12, 14, 16
11	Metals, insulators and semiconductors	6.5	4, 13, 17

Questions are drafted for the gaps in the past papers (not from any past paper). Section 2.3 (X-ray diffraction methods) is outside the Module I scope, so no question uses it.